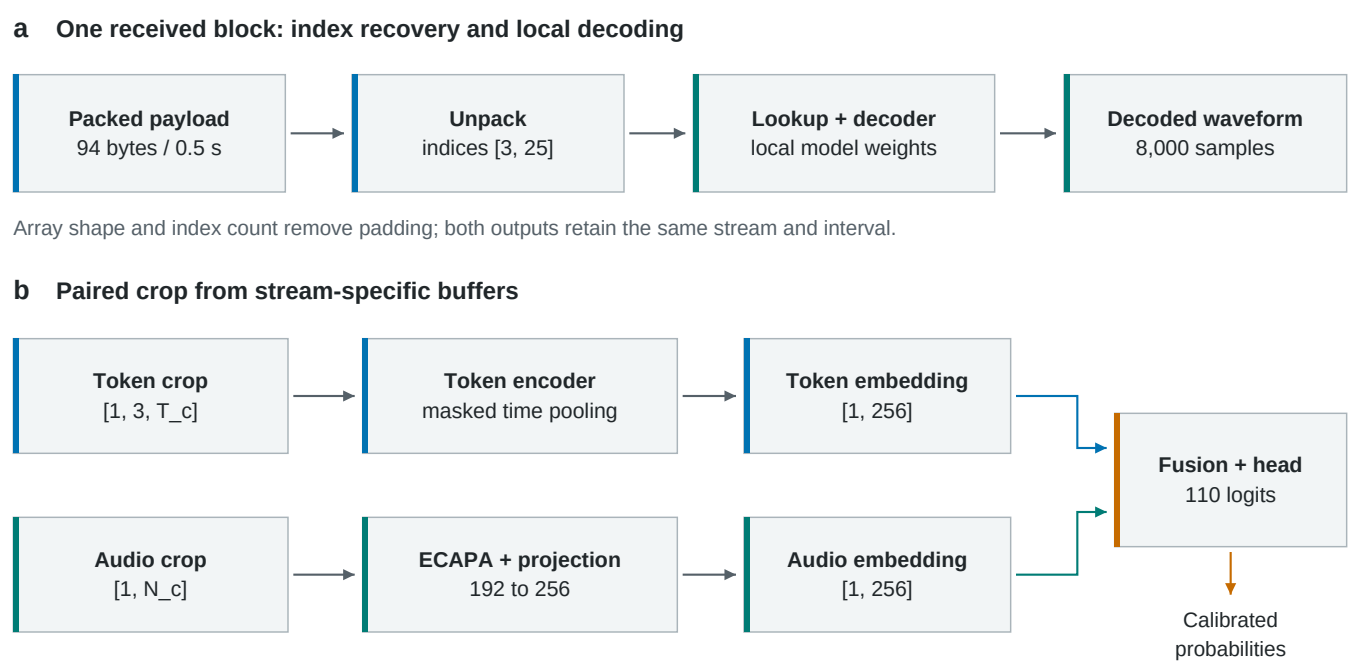
a
One received block: index recovery and local decoding
Packed payload
94 bytes / 0.5 s
Unpack
indices [3, 25]
Lookup + decoder
local model weights
Decoded waveform
8,000 samples
Array shape and index count remove padding; both outputs retain the same stream and interval.
b
Paired crop from stream-specific buffers
Token crop
[1, 3, T_c]
Token encoder
masked time pooling
Token embedding
[1, 256]
Audio crop
[1, N_c]
ECAPA + projection
192 to 256
Audio embedding
[1, 256]
Fusion + head
110 logits
Calibrated
probabilities
N_c: actual cropped audio length (up to 48,000). T_c: matched code length from the cache or buffer.